Calculation of Energy/Forces

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Using modern supercomputers, one can realistically simulate upto 10^6 molecules.

However, in typical experiments, the system sizes involve 10^{23} molecules (1 mole)

As system size increases, the fraction of molecules present on the surface progressively decreases

if the size of a system is "L"
then surface area $\sim L^2$
volume $\sim L^3$

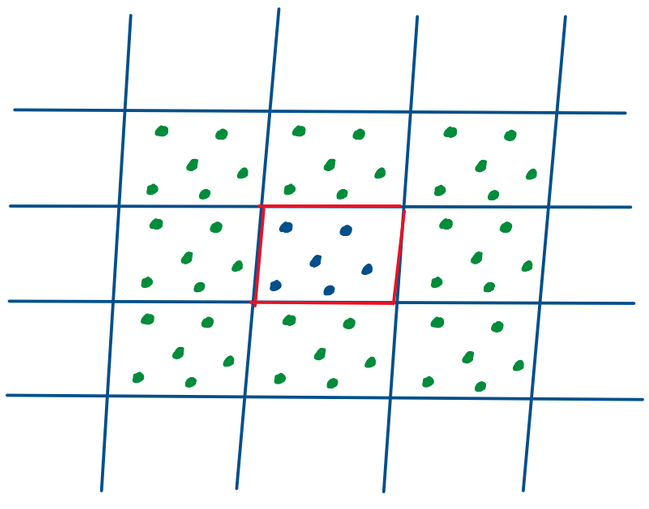
$\Rightarrow$ Fraction of molecules in the surface $\sim \frac{L^2}{L^3} = \frac{1}{L} \sim \frac{1}{N^{1/3}}$

if $N \sim 10^{23}$ fraction of surface molecules $(10^{-7} - 10^{-8})$
if $N \sim 10^6$ " 10^{-2}

$\Rightarrow$ properties of a system with 10^6 molecules will differ from those with 10^{23} molecules even if we use the correct molecular model

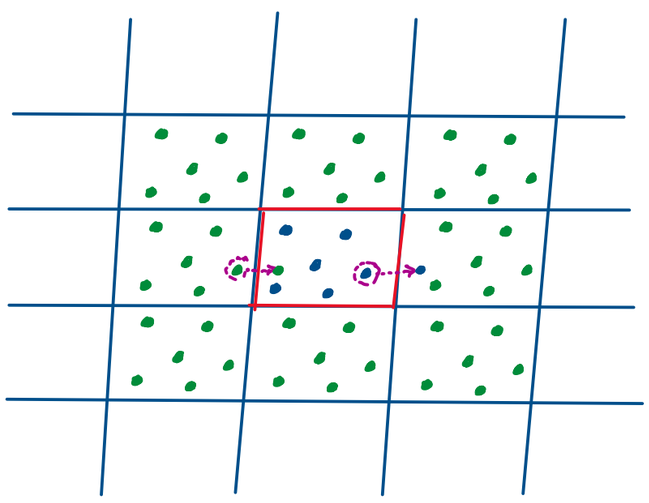
To overcome this problem, one applies "Periodic boundary conditions" on the systems

Periodic Boundary Conditions (PBC)



In PBC, the system is replicated in all three dimensions. As a result, every molecule is surrounded by other molecules and all surface effects are eliminated.
While computing interaction of a molecule, one considers not only the original molecules in the the simulation box, but also interactions with all the periodic images

In a molecular simulation, molecules are constantly moving. This movement is mirrored by all the periodic images.
Since the boundaries of our simulation box are imaginary, a molecule can leave the box.
When this happens, a corresponding image of the molecule enters the box.
So we only track positions of those image that lie in the original simulation box.



Implementation
Let $\{r_x, r_y, r_z\}$ be position of the molecule
$L \rightarrow$ length of the cubic simulation box
Boundaries of the simulation box are located at $-\frac{L}{2}$ and $\frac{L}{2}$

Pseudo code

$$r_x \leftarrow r_x - L * \text{ANINT}\left(\frac{r_x}{L}\right)$$

$$rx \leftarrow rx - L * \text{ANINT}\left(\frac{rx}{L}\right)$$

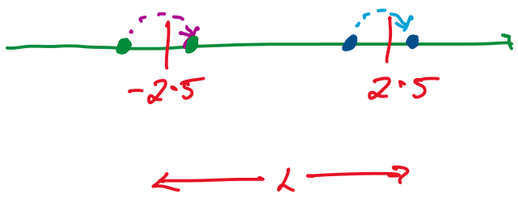
$$ry \leftarrow ry - L * \text{ANINT}\left(\frac{ry}{L}\right)$$

$$rz \leftarrow rz - L * \text{ANINT}\left(\frac{rz}{L}\right)$$

ANINT :- Fortran function
Returns the nearest integer
e.g. ANINT(2.3) = 2.0
ANINT(5.7) = 6.0

Example in 1D

$L = 5$



$x^{(old)} = 2.0$

$x^{(new)} = 3.0$

apply PBC

$$x^{(new)} = 3.0 - 5.0\ \text{ANINT}\left(\frac{3.0}{5.0}\right)$$

$$= 3.0 - 5.0 * 1.0$$

$$= -2.0$$

Calculation of energy due to intermolecular interaction for a system under PBC

Let $u(r)$ be the potential energy between a pair of atoms separated by a distance r.

Consider a system with "N" atoms in a cubic simulation box of length L

Potential energy $= U = \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ \neq i}}^{\infty} u(r_{ij})$

r_{ij} ≡ distance between atoms i and j

The upper limit is ∞ since we have applied PBC to the system

$$U = \frac{1}{2} \sum_{i=1}^{N} u_i$$

$$u_i = \sum_{\substack{j=1 \\ \neq i}}^{\infty} u(r_{ij})$$

Since it is not possible to perform a summation over infinite pairs, we divide the above summation into two sets,

(1) A set of distances that are smaller than a pre-specified cut-off distance, r_c
i.e. $r_{ij} \leq r_c$

2) Rest of the pairs i.e. $r_{ij} > r_c$

$$\therefore u_i = \sum_{\{j;\, r_{ij} \leq r_c\}} u(r_{ij}) + \sum_{\{j;\, r_{ij} > r_c\}} u(r_{ij})$$

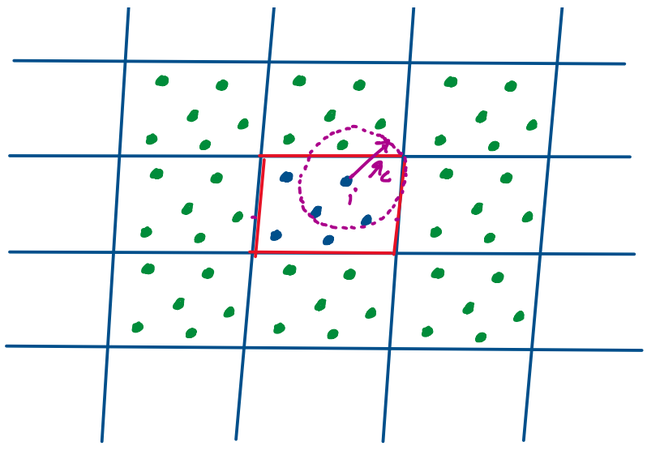
$$= u_i^{(SR)} + u_i^{(LR)}$$

$u_i^{(SR)}$: short range interaction (SR); contains finite number of terms computed explicitly in a simulation

$u_i^{(LR)}$: long range interaction (LR); contains infinite number of terms computed analytically with a suitable approximation

$$u_i^{(SR)} = \sum_{\{r_{ij} \leq r_c\}}^{j} u(r_{ij})$$

One has to identify all the nearest neighbours for a system subjected to PBC as shown in the figure below.



However, we only store coordinates of atoms in the main simulation box.
Hence, we have to compute distance to the closest (nearest) image
This is done as follows.

Let $\vec{r}_i$ and $\vec{r}_j$ be the coordinates of the atoms

1) $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$

$\vec{r}_{ij} \equiv \{x_{ij}, y_{ij}, z_{ij}\}$

2)
$$x_{ij} \leftarrow x_{ij} - L * \text{ANINT}\left(\frac{x_{ij}}{L}\right)$$
$$y_{ij} \leftarrow y_{ij} - L * \text{ANINT}\left(\frac{y_{ij}}{L}\right)$$
$$z_{ij} \leftarrow z_{ij} - L * \text{ANINT}\left(\frac{z_{ij}}{L}\right)$$

vector from "i" to the closest image of "j".

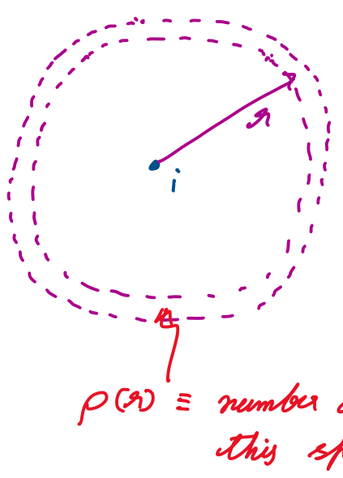
3) Compute $r_{ij} = \sqrt{x_{ij}^2 + y_{ij}^2 + z_{ij}^2}$

if $r_{ij} \leq r_c$ then

· compute $u(r_{ij})$ and add it to $u_i^{(SR)}$

In almost all simulations, one only includes interaction with the closest image in $u_i^{(SR)}$.
The interactions with the rest of the images is included in $u_i^{(LR)}$
This convention of only including the nearest image in our calculation of $u_i^{(SR)}$ is called "Minimum Image Convention"

Since atoms interact with each other, the position of any neighboring atom "j" from a given atom "i" is correlated.
At long distances ($r_{ij} > r_c$), the interaction is weak and we can assume that atoms become uniformly distributed.



At long distance $\rho(r) \simeq \rho \equiv$ bulk density of the system (since particle positions are uncorrelated)

$$\equiv \frac{N}{V}$$

$$u_i^{(LR)} = \int_{r_c}^{\infty} 4\pi r^2 u(r) \rho \, dr$$
$$= 4\pi\rho \int_{r_c}^{\infty} r^2 u(r) \, dr$$

$$\therefore \quad U = \frac{1}{2} \underbrace{\sum_{i=1}^{N} u_i^{(SR)}}_{U^{(SR)}} + \frac{N}{2} 4\pi\rho \int_{r_c}^{\infty} r^2 u(r) \, dr$$
$$= U^{SR} + \frac{2\pi N^2}{V} \int_{r_c}^{\infty} r^2 u(r) \, dr$$

To compute force on atom i" we take gradient of U w.r.t. $\vec{r}_i$

$$\vec{f}_i = \nabla_{\vec{r}_i} U = \nabla_{\vec{r}_i} U^{(LR)}$$

The above method makes an implicit assumption that the integral in $u_i^{(LR)}$ is convergent. This need not be the case always.

Let $u(r) \sim \frac{C}{r^n}$

$$\therefore U^{LR} \sim \int_{r_c}^{\infty} \frac{r^2}{r^n} dr = \int_{r_c}^{\infty} \frac{1}{r^{n-2}} dr$$

The above integral is convergent only when $\boxed{n > 3}$

e.g Lennard-Jones potential $u(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$

However for electrostatic interactions $u(r) \sim \frac{1}{r}$

$$\therefore U^{LR} \sim \int_{r_c}^{\infty} r\, dr \equiv \text{not convergent}$$

In such situations one has to use specialized mathematical techniques.
One such technique is called the "Ewald Summation"